

2.1 Memory map and peripheral addressing

The ARM Cortex-M3 processor has a single 4 GB address space. The following table shows how this space is used on the LPC176x/5x.

Table 3. LPC176x/5x memory usage and details

Address range	General Use	Address range details and description	
0x0000 0000 to 0x1FFF FFFF	On-chip non-volatile memory	0x0000 0000 - 0x0007 FFFF	For devices with 512 kB of flash memory.
		0x0000 0000 - 0x0003 FFFF	For devices with 256 kB of flash memory.
		0x0000 0000 - 0x0001 FFFF	For devices with 128 kB of flash memory.
		0x0000 0000 - 0x0000 FFFF	For devices with 64 kB of flash memory.
		0x0000 0000 - 0x0000 7FFF	For devices with 32 kB of flash memory.
	On-chip SRAM	0x1000 0000 - 0x1000 7FFF	For devices with 32 kB of local SRAM.
		0x1000 0000 - 0x1000 3FFF	For devices with 16 kB of local SRAM.
		0x1000 0000 - 0x1000 1FFF	For devices with 8 kB of local SRAM.
	Boot ROM	0x1FFF 0000 - 0x1FFF 1FFF	8 kB Boot ROM with flash services.
	0x2000 0000 to 0x3FFF FFFF	On-chip SRAM (typically used for peripheral data)	0x2007 C000 - 0x2007 FFFF
0x2008 0000 - 0x2008 3FFF			AHB SRAM - bank 1 (16 kB), present on devices with 64 kB of total SRAM.
GPIO		0x2009 C000 - 0x2009 FFFF	GPIO.
0x4000 0000 to 0x5FFF FFFF	APB Peripherals	0x4000 0000 - 0x4007 FFFF	APB0 Peripherals, up to 32 peripheral blocks, 16 kB each.
		0x4008 0000 - 0x400F FFFF	APB1 Peripherals, up to 32 peripheral blocks, 16 kB each.
	AHB peripherals	0x5000 0000 - 0x501F FFFF	DMA Controller, Ethernet interface, and USB interface.
0xE000 0000 to 0xE00F FFFF	Cortex-M3 Private Peripheral Bus	0xE000 0000 - 0xE00F FFFF	Cortex-M3 related functions, includes the NVIC and System Tick Timer.

2.2 Memory maps

The LPC176x/5x incorporates several distinct memory regions, shown in the following figures. [Figure 3](#) shows the overall map of the entire address space from the user program viewpoint following reset. The interrupt vector area supports address remapping, which is described later in this section.

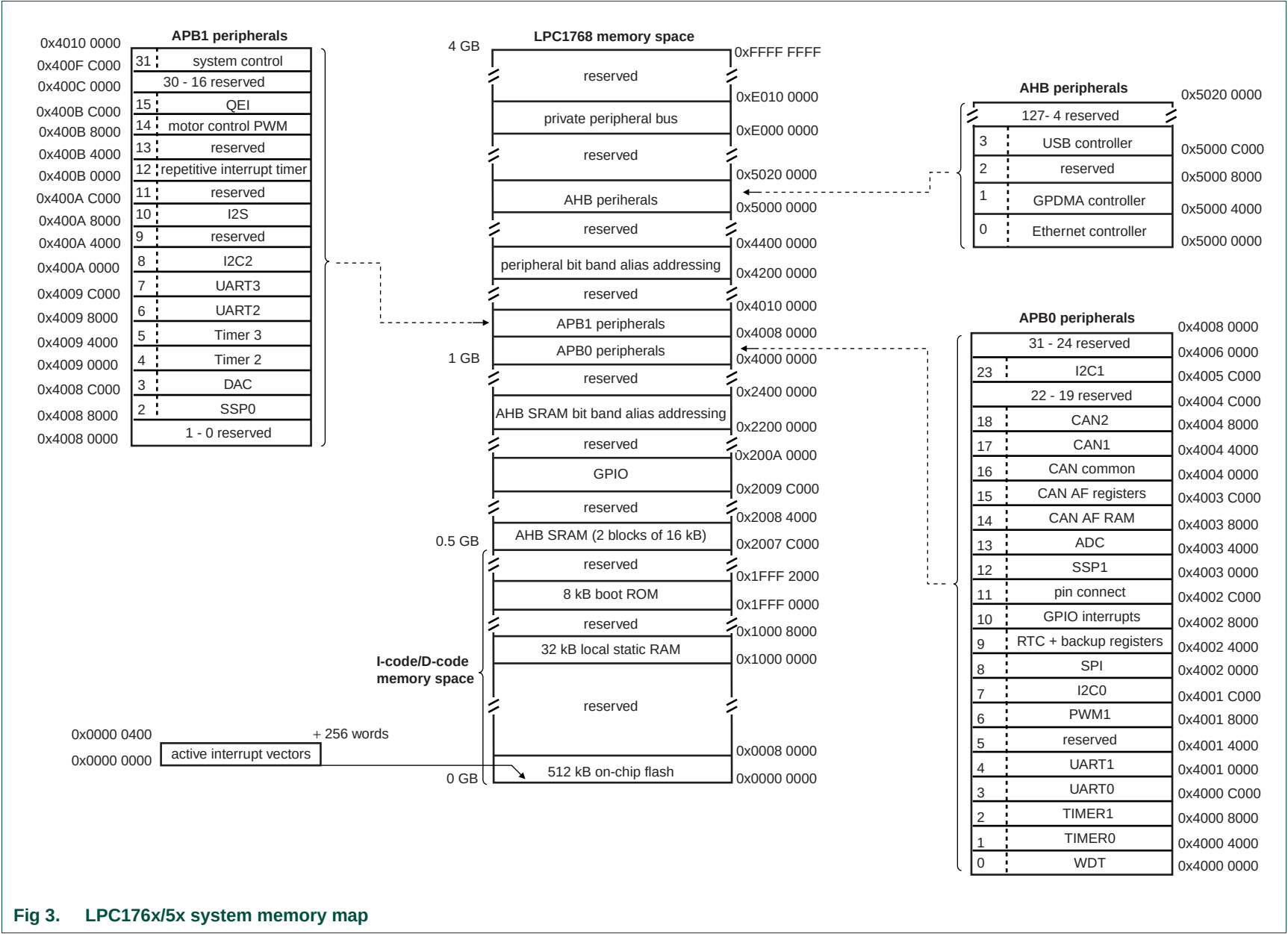


Fig 3. LPC176x/5x system memory map

[Figure 3](#) and [Table 4](#) show different views of the peripheral address space. The AHB peripheral area is 2 megabyte in size, and is divided to allow for up to 128 peripherals. The APB peripheral area is 1 megabyte in size and is divided to allow for up to 64 peripherals. Each peripheral of either type is allocated 16 kilobytes of space. This allows simplifying the address decoding for each peripheral.

All peripheral register addresses are word aligned (to 32-bit boundaries) regardless of their size. This eliminates the need for byte lane mapping hardware that would be required to allow byte (8-bit) or half-word (16-bit) accesses to occur at smaller boundaries. An implication of this is that word and half-word registers must be accessed all at once. For example, it is not possible to read or write the upper byte of a word register separately.

2.3 APB peripheral addresses

The following table shows the APB0/1 address maps. No APB peripheral uses all of the 16 kB space allocated to it. Typically each device's registers are "aliased" or repeated at multiple locations within each 16 kB range.

Table 4. APB0 peripherals and base addresses

APB0 peripheral	Base address	Peripheral name
0	0x4000 0000	Watchdog Timer
1	0x4000 4000	Timer 0
2	0x4000 8000	Timer 1
3	0x4000 C000	UART0
4	0x4001 0000	UART1
5	0x4001 4000	reserved
6	0x4001 8000	PWM1
7	0x4001 C000	I ² C0
8	0x4002 0000	SPI
9	0x4002 4000	RTC
10	0x4002 8000	GPIO interrupts
11	0x4002 C000	Pin Connect Block
12	0x4003 0000	SSP1
13	0x4003 4000	ADC
14	0x4003 8000	CAN Acceptance Filter RAM
15	0x4003 C000	CAN Acceptance Filter Registers
16	0x4004 0000	CAN Common Registers
17	0x4004 4000	CAN Controller 1
18	0x4004 8000	CAN Controller 2
19 to 22	0x4004 C000 to 0x4005 8000	reserved
23	0x4005 C000	I ² C1
24 to 31	0x4006 0000 to 0x4007 C000	reserved

Table 5. APB1 peripherals and base addresses

APB1 peripheral	Base address	Peripheral name
0	0x4008 0000	reserved
1	0x4008 4000	reserved
2	0x4008 8000	SSP0
3	0x4008 C000	DAC
4	0x4009 0000	Timer 2
5	0x4009 4000	Timer 3
6	0x4009 8000	UART2
7	0x4009 C000	UART3
8	0x400A 0000	I ² C2
9	0x400A 4000	reserved
10	0x400A 8000	I ² S
11	0x400A C000	reserved
12	0x400B 0000	Repetitive interrupt timer
13	0x400B 4000	reserved
14	0x400B 8000	Motor control PWM
15	0x400B C000	Quadrature Encoder Interface
16 to 30	0x400C 0000 to 0x400F 8000	reserved
31	0x400F C000	System control

2.4 Memory re-mapping

The Cortex-M3 incorporates a mechanism that allows remapping the interrupt vector table to alternate locations in the memory map. This is controlled via the Vector Table Offset Register contained in the Cortex-M3. Refer to [Section 6.4](#) and [Section 34.4.3.5](#) of the Cortex-M3 User Guide appended to this manual for details of the Vector Table Offset feature.

Boot ROM re-mapping

Following a hardware reset, the Boot ROM is temporarily mapped to address 0. This is normally transparent to the user. However, if execution is halted immediately after reset by a debugger, it should correct the mapping for the user. See [Section 33.6](#).

2.5 AHB arbitration

The Multilayer AHB Matrix arbitrates between several masters. By default, the Cortex-M3 D-code bus has the highest priority, followed by the I-Code bus. All other masters share a lower priority.

2.6 Bus fault exceptions

The LPC176x/5x generates Bus Fault exception if an access is attempted for an address that is in a reserved or unassigned address region. The regions are areas of the memory map that are not implemented for a specific derivative. These include all spaces marked “reserved” in [Figure 3](#).

For these areas, both attempted data access and instruction fetch generate an exception. In addition, a Bus Fault exception is generated for any instruction fetch that maps to an AHB or APB peripheral address.

Within the address space of an existing APB peripheral, an exception is not generated in response to an access to an undefined address. Address decoding within each peripheral is limited to that needed to distinguish defined registers within the peripheral itself. For example, an access to address 0x4000 D000 (an undefined address within the UART0 space) may result in an access to the register defined at address 0x4000 C000. Details of such address aliasing within a peripheral space are not defined in the LPC176x/5x documentation and are not a supported feature.

If software executes a write directly to the flash memory, the flash accelerator will generate a Bus Fault exception. Flash programming must be accomplished by using the specified flash programming interface provided by the Boot Code.

Note that the Cortex-M3 core stores the exception flag along with the associated instruction in the pipeline and processes the exception only if an attempt is made to execute the instruction fetched from the disallowed address. This prevents accidental aborts that could be caused by prefetches that occur when code is executed very near a memory boundary.